



ETL TESTING

MASTERCLASS

Introduction to ETL Testing

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One of the key elements for a successful Data Warehouse is the ability of the test team –

o Plan | o Design | o Execute effective tests

• ETL Testing helps identify issues related to:

- o 🕒 Data/time consistency
- o ✅ Data quality
- o ❌ Failures in Extract–Transform–Load (ETL) process

The primary focus should be on testing the ETL process, including:

- o Loading of all required rows
- o Correct execution of all transformations
- o Successful completion of data cleaning

Data Warehouse projects mainly focus on:

- o Moving data from one system (source) to another system (target)
- o To enable reporting and analysis

Source data can come from:

- o Databases of applications | o Flat files | o Other application outputs

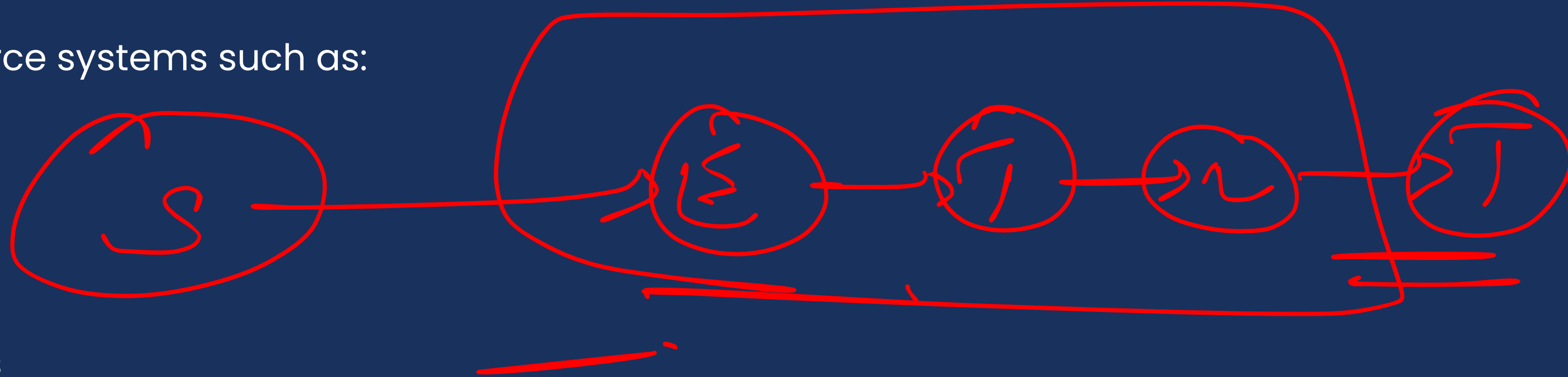
ETL – Extract, Transform, Load

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It is the process of loading data from source systems to the Data Warehouse.

● 2.1 Extraction

- Data is pulled from source systems such as:
 - Databases
 - Flat files
 - XML files
 - Excel files
 - CSV files
 - Any other file systems



● 2.2 Transformation

- Data is transformed into new formats according to business rules
- Business logic is applied at this stage
- This is often the most complex part of a Data Warehouse project

● 2.3 Loading

- Transformed data is loaded into the Data Warehouse / Target system

Test Strategy for ETL Testing

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These are common ETL Testing strategies:

1.  Data Completeness
2.  Data Transformation
3.  Data Quality
4.  Initial / Full Load Testing
5.  Incremental / Delta Load Testing
6.  Presentation Layer (BI) Testing
7.  Job Dependency Testing

Types of Load

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1. ● Full Load (Bulk Load / First Time Load / Fresh Load).

- All data loaded into target for the first time

2. ● Incremental Load (Refresh Load / Delta Load).

- Only modified data loaded to target after full load
- Changes captured using:
 - 1.Created Date / Modified Date
 - 2.Compared against last job run date

Data Cleaning (Data Scrubbing).

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 **Goal: Remove incorrect data and improve confidence in data.**

Data Cleaning = Removing irrelevant, redundant, incorrect and incomplete data.

-  Irrelevant → Delete data not required for business anymore
-  Redundant → Remove duplicate data
-  Incorrect → Update wrong values with correct values
-  Incomplete → Fill missing values with full information

Common data issues to clean before loading into DWH

Data Masking

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 **Data Masking = Hiding / Masking / Encrypting confidential or sensitive data.**

 **Types:** Static Masking | Dynamic Masking

 **Techniques:** Replacing subset of data with:

- Dummy characters
- Dummy numbers
- Random numbers
- Dummy special characters
- Shuffling characters or numbers

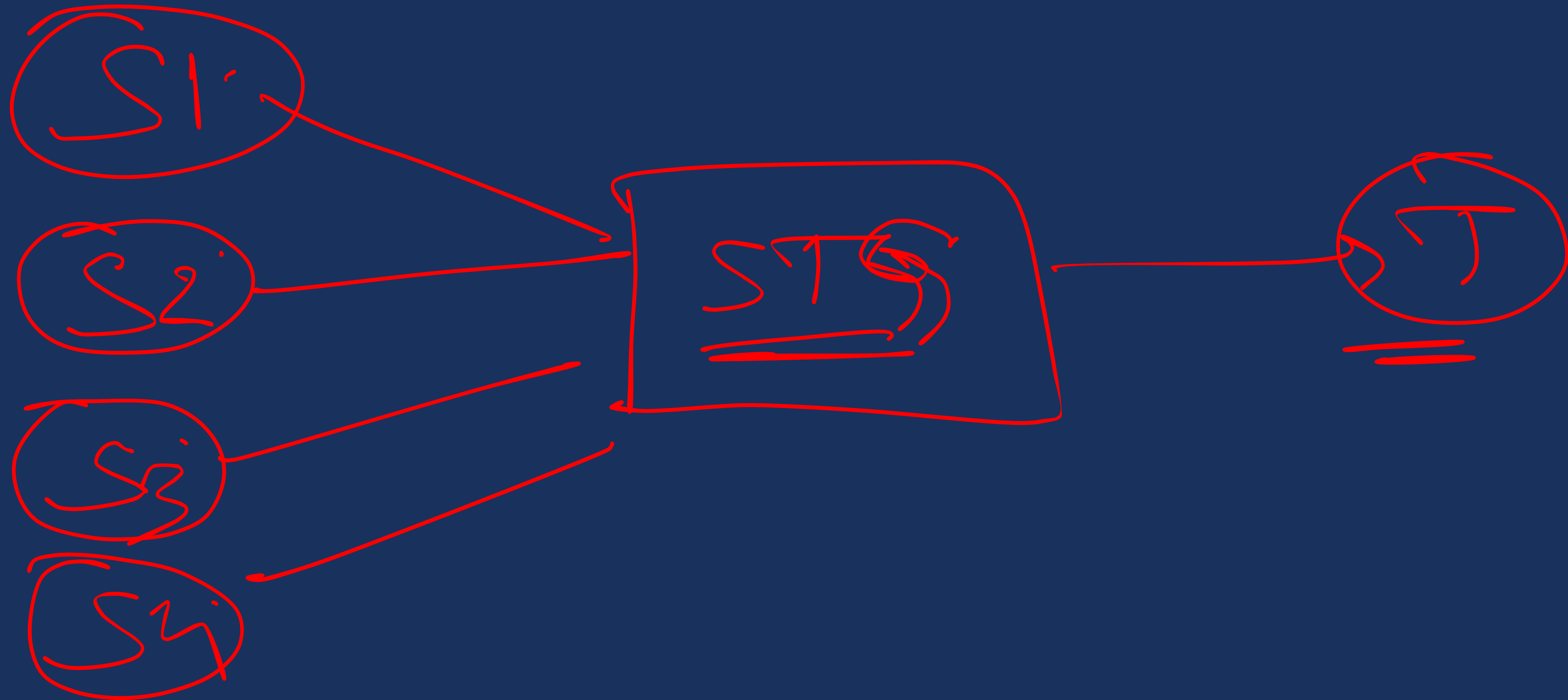
NOTE:

ETL Tester checks whether masking is applied correctly,
but does not test the internal masking algorithms / logic.

Data Merging

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Integrating data from multiple operational sources into a single pipeline.



ODSOperational Data Store

- Integrated data from different business sources with defined rules
- Gets data directly from transactional databases
- Stores limited period of historical data
- A big company can have multiple ODS (varies project to project)

DWHWarehouse

- Enterprise Data Warehouse = Central system for:
 - Reporting
 - Data analysis
- Stores integrated data from one or more sources
- Core component of Business Intelligence (BI)

DMData Mart

- Subset of Data Warehouse
- Oriented to a specific business line / team
- Example: Sales Mart, Finance Mart, HR Mart

Data Retention – Archiving & Purging

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 **Data Retention = How long data must be kept in the database.**

 Why Data Retention Policy?

 Purging

 Archiving

 Selective Archiving or Purging

ETL Testing Life Cycle / Process

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Similar to other testing types, ETL Testing has its own phases:

1. Requirement Analysis
2. Test Planning
3. Test Design
4. Test Execution
5. Defect Retesting / Reporting
6. Test Closure / Sign-off

ETL Test Scenarios – Common ETL Test Checks:

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1. Table Structure Verification ✓
2. Constraint Check ✓
3. Index Check ✓
4. Source Data Validation ✓
5. Data Count Check ✓ S → T
6. Data Comparison ✓ S T
7. Duplicate Data Validation ✓
8. PK / FK Validation ✓
9. Null Check ✓
10. Data Precision Check ✓
11. Date Format Check ✓
12. Business Transformation Rules Validation ✓

Active & Passive Transformations

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● Active Transformation

- Output record count may be different from input
- Example: Filter – salary > 30000, age between 18 and 30

● Passive Transformation

- Output record count = Input record count
- Example: Concatenate First Name + Last Name

Types of Transformations (Examples). 13

1. Filter Transformation (Active)
2. Expression Transformation (Passive)
3. Aggregator Transformation (Active)
4. Joiner Transformation (Active)
5. Union Transformation (Active)
6. Sorter Transformation (Passive)
7. Rank Transformation
8. Sequence Generator Transformation
9. Router Transformation
10. Lookup Transformation

ETL Documents

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- 📄 Data Model & Mapping Sheet
- 📄 ETL Test Cases
- 📄 Business Requirement Document (BRD)

Mapping document includes:

- Column mapping between Source & Target
- Data types & length
- Transformation logic per column
- ETL jobs information

Source & target DB schema should be handy.

ETL Test Cases – Examples

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 Source Schema Validation

 Target Schema Validation

 Mapping Test Cases

Common ETL Bugs

- Table structure issues ✓
- Data issues in source ✓
- Count mismatch (source vs target) ✓
- Data mismatch (source vs target) ✓
- Duplicate data loaded ✓
- Trim / NULL handling issues ✓
- Data precision issues ✓
- Data / Date format issues ✓
- Business logic issues ~~✓~~
- Performance issues ~~✓~~

~~Data~~



Best Practices in ETL Defect Logging

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When raising ETL defects, include:

- Clear summary
- Test data used
- Relevant SQL queries
- Detailed description of the issue
- Screenshot / evidence

Key Responsibilities of an ETL Tester

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- Understand requirements
- Raise clarifications
- Prepare test plan
- Prepare test cases with SQL queries
- Review test cases & queries
- Execute test cases
- Raise defects
- Attend defect triage calls
- Track defects
- Share regular status
- Provide sign-off
- Prepare documents as per team needs

Data Completeness Test Strategy.

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Data completeness tests verify that – *Records / Fields / Full content*

Key checks:

1. Record Count Check
2. Duplicate Record Check
3. Column Mapping Check
4. Field Content Check

Challenges in DWH Testing

- Very high data volume
- Data from heterogeneous sources
- Poor data quality at source → more QA effort
- Complex business logic (joins, subqueries, set operations)
- Limited or unavailable test data
- No proper documentation
- Need to attend multiple team meetings: Dev | Architect | Support
- Need to coordinate with many teams to get data / info
- Need to create own documentation with available information

Slowly Changing Dimension (SCD).

20.1

SCD = Dimension that stores **current + historical data** over time.

Used where attribute values change over time.

Types of SCD:

SCD Type 1 – Overwrite Data

SCD Type 2 – Add new Record

SCD Type 3 – Add new Column

Slowly Changing Dimension (SCD).

20.2

SCD Type 1 – Overwrite Data

Cust_id	Cust_name	City
101	Ram	Mumbai
102	Krishna	Bangalore

Before



Cust_id	Cust_name	City
101	Ram	Bangalore
102	Krishna	Bangalore

After

history?

Slowly Changing Dimension (SCD).

SCD Type 2 – Add new Record

Before

Surrogate_Key	Customer_ID	Customer_Name	City	Start_Date	End_Date	Current_Flag
1	101	Ravi	Mumbai	1/1/2024	NULL	Y

After

Surrogate_Key	Customer_ID	Customer_Name	City	Start_Date	End_Date	Current_Flag
1	101	Ravi	Mumbai	1/1/2024	1/5/2025	N
2	101	Ravi	Bangalore	2/5/2025	NULL	Y

3 101 Ravi Hyd 1/1/2024 30/12/2024 Y

Slowly Changing Dimension (SCD).

SCD Type 3 - Add new Column

Before

Customer_ID	Customer_Name	City_Current	City_Previous
101	Ravi	Mumbai	NULL

After

Customer_ID	Customer_Name	City_Current	City_Previous
101	Ravi	Bangalore	Mumbai

101 Row type Bangalore

Surrogate Key

- Acts like a Primary Key
- Used to maintain uniqueness in SCD2 tables

Surrogate_Key	Customer_ID	Customer_Name	City	Start_Date	End_Date	Current_Flag
1	101	Ravi	Mumbai	1/1/2024	1/5/2025	N
2	101	Ravi	Bangalore	2/5/2025	NULL	Y

Numeric

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Job Dependency Testing

-  **Parallel Job** – Multiple jobs run at the same time
-  **Sequential Job** – One job depends on successful completion of another

Job Scheduler comes into picture to:

- Schedule multiple jobs
- Maintain job sequence



THANKS

Notes – Links in Description

